

Model-Based Cooperative Navigation for a Group of Flying Robots

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In this article, a model-based cooperative navigation system for a group of quadrotors, while no absolute position data is accessible, is proposed. The growth of position errors in a GNSS-denied condition is such that the navigation methods are not reliable. In the proposed method, instead of the conventional inertial navigation equations, a different dynamic equation based on drag forces is used. Together with the relative position measurements between agents, the acceleration measurements on the plane of the rotors of at least one of the agents are also used in the observation process. For evaluation of the proposed method, an algorithm-in-loop 6DOF simulation environment is developed. The results illustrate that the growth rate of navigation errors is significantly reduced; 97% in the horizontal plane and 40% in the vertical one. In short, integrating the proposed model-based and cooperative navigation methods is more effective than using each of them separately.

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I. INTRODUCTION

With the advent of a new generation of mobile robots (MR), an idea called “cooperative robotics” has recently emerged to ease the achievement of complex missions. A group of flying robots (FRs) is more robust than a single one, in which more complex missions can be performed [1]. Most of the prevailing navigation approaches are devoted to the optimal fusion of the measurements of proprioceptive and exteroceptive instruments for measuring vehicle motion and surrounding environment signals. To resolve accumulated errors of inertial measurement unit (IMU), position fixing methods (i.e., GNSS) are fused with the dead reckoning (DR) ones. However, GNSS signals may be unavailable for a variety of reasons.

In this regard, Kurazume et al. proposed “cooperative positioning” in 1994 [2]. Later, a method called “collective localization” without using any external source was presented [3]. Using linear observability analysis, it was found that the system is not observable unless at least one of the robots accesses the absolute position. Rekleitis et al. [4] investigated the effects of different factors like relative range and bearing measurements and the number of agents on the performance of CN. In this regard, Rekleitis and Roumeliotis [5] and [6] followed analytical solutions for the odometry errors. For the first time, Martinelli and Siegwart [7] presented a nonlinear observability analysis method based on Lie algebra for two MRs. Later, a closed-form solution for steady-state position error covariance in [8] was presented.

CN was first introduced by Sharma and Taylor in which, by nonlinear observability analysis, they verified that the IMU biases are observable [9]. Next, they presented a distributed CN by measuring only the relative bearing angle between the agents [10]. Moreover, a consistent CN for multiple autonomous underwater vehicles (AUVs) was proposed in [11]. Afterward, Gao and Chitre [12] presented a CN for two AUVs using range-only measurements. To analyze the nonlinear observability of the bearing-only CN, a graph-based method was proposed in [13]. At the same time, a centralized cooperative localization method, based on relative measurements between the agents and landmarks with a known position, was presented in a GNSS-denied condition [14]; however, Goel et al. [15] used a UWB module in each agent as a relative sensor. Later, Chakraborty et al. [16] demonstrated that each group can localize itself using only two known landmarks and relative bearing-only observations. Russell et al. [17] solved the problem of localizing a GPS-denied UAV (agent B) with the assistance of a GPS-enabled UAV (agent A), cooperatively.

In references that investigate model-based inertial navigation, the two terms “model aided” and “model-based” are well known. In “model aided” methods a precise dynamic model alongside the INS is used. Koifman and Bar-Itzhack [18] are well-known pioneers in this field. In their work, a complete model of the dynamics of a fixed-wing drone is presented as an alternative to the GPS, in which the error of the coefficients and parameters in the model during the

flight is estimated and finally compensated by an EKF. In other words, in this method, there are two-way corrections between the INS and the dynamic model. All subsequent citations in the field (i.e., [19]–[23]) provide minor contributions in the field.

References that use the “model-based” term employ a model such as an algebraic constraint or a differential constraint. Some authors introduce a drag force model (DFM) for use in dynamic equations as well as in the observation model. In the field tests, it was found that the DF from rotor blades is a linear function of the body velocity in the plane of the rotors [24]; where drag of rotor blades can be measured by accelerometers and provide an improvement in attitude estimates, as well. In this regard, the main citations are [25] and [26].

According to the reviews, the CN of flying robots in which no agent has access to absolute position information still demands more effort. In this regard, an analytical solution was derived by Faghihinia et al. [27]. The author demonstrated that the position error covariance does not depend on flight trajectory and noise characteristics of the relative position measurements, while it depends on the number of the agents in the group and the noise covariance of the accelerometers. To enhance the CN in [27], in this article, a drag force-based (DFB) approach is proposed.

The rest of this article is organized as follows: In Section II-A, the equations related to a single agent model-based navigation are derived. Section II-B describes cooperative navigation and related derivations, and Section II-C demonstrates the model-based cooperative navigation algorithm. Next, the results of the simulations, developed in MATLAB, and the evaluation of the proposed method are presented in Section III. Finally, Section IV concludes this article.

II. METHODS

A. Single-Agent Model-Based Navigation

In this article, the group of FRs consists of “ N ” quadrotors that operate under a GNSS-denied condition and utilize an EKF to perform the CN algorithm. It is assumed that each quadrotor is equipped with an IMU to perform the DR process (propagation) and relative sensors to measure the interagent relative range and bearing angles. Since the operating conditions are at low speed and range, the North East Down (NED) frame is adopted as a reference frame (see Fig. 1).

In the field of model-based navigation of multirotors, several papers, such as [24]–[26], [28], benefit from a DF model in their estimator. Various experiments have shown that the accelerometers in both x - and y -directions in the body frame, measure the acceleration due to the rotor’s drag where it profits from a linear trend with velocities

$$\begin{aligned} d_x &= \mu v_x \\ d_y &= \mu v_y \end{aligned} \quad (1)$$

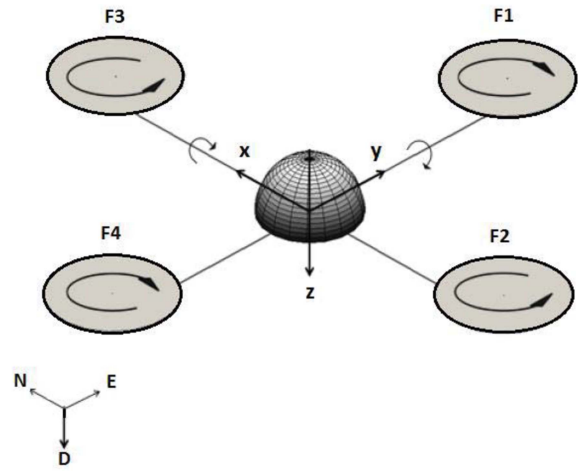


Fig. 1. NED reference and body coordinate.

where μ is a positive constant number, v_x and v_y denote the velocity components, and d_x , d_y stand for the DF components, all in the body coordinate system, which comprises inductive and rotor profile drag. The DF coefficient is modeled by the following equation [24]:

$$\mu = \lambda \sum_{i=1}^N \omega_i \quad (2)$$

where the parameter λ is a constant number and ω_i stands for the rotational speed of the i th rotor. However, nominal flight conditions for a quadrotor dictates that the sum of rotors’ speed is nearly constant, which results in μ to be constant [26]. In this method, the accelerations measured on the rotor plane are used as the main observations in the EKF update step. In the DR process, a_x and a_y , which are the body accelerations in x - and y -directions are calculated through the following relation:

$$\begin{aligned} a_x &= -\frac{\mu}{m} v_x \\ a_y &= -\frac{\mu}{m} v_y. \end{aligned} \quad (3)$$

Fig. 2 shows the flowchart of a navigation algorithm, based on a DF enhanced model, for a single agent. In this algorithm, only the IMU data is used during the prediction and update phases while not exploited any absolute position data.

As Fig. 2 illustrates, in the update step, a_x and a_y in (3) are also used as an observation model. Hence, the body velocity components can be directly observed by measuring the acceleration. On the other hand, due to the effect of the roll and pitch angle on the rate of body velocities (due to the terms of the gravitational acceleration in velocity differential equations), these two angles are also observable by measuring the body acceleration. By transferring the velocity vector from the body to the NED coordinate system, the position of the robot can also be determined. It should be noted that in this algorithm there is no observation of the position elements, so to obtain an accurate position, there must be at least the initial absolute position with sufficient

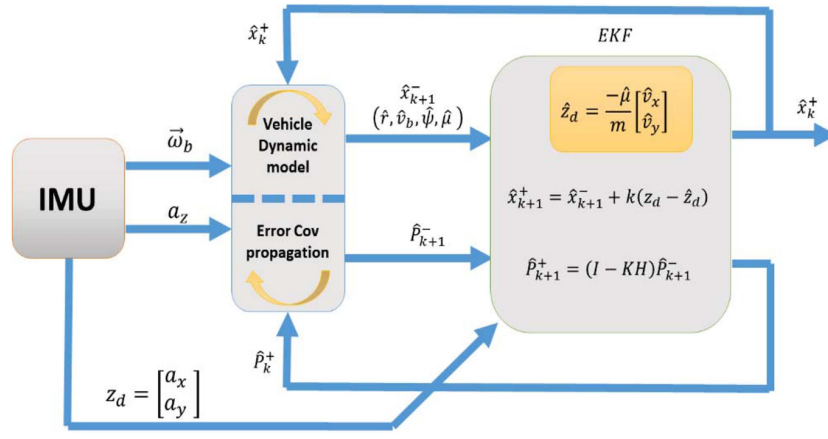


Fig. 2. Block diagram of the DF enhanced MB navigation for a single agent.

accuracy; otherwise, the initial position error cannot be corrected by acceleration observations. In the following, MB navigation equations are developed using the DFM. Since μ is not known accurately, it is augmented in the estimation vector as a new state variable in EKF. Therefore, the estimation state vector is defined as follows:

$$x = [r \ v_b \ \chi \ \mu] \quad (4)$$

where χ denotes the attitude angles of the body relative to the NED reference frame, v_b is the body velocity vector in the body coordinate, and r signifies the position vector in the NED coordinate system. Therefore, the 6DOF equations of motion using the DFM are given as follows:

$$\begin{aligned} \dot{r}_e &= C_b^e v_b \\ \dot{v}_b &= \begin{bmatrix} -\frac{\mu}{m} v_x \\ -\frac{\mu}{m} v_y \\ \frac{a_z}{m} \end{bmatrix} - \omega_b \times v_b + C_e^b g^e \\ \dot{\chi} &= R_\chi \omega_b \\ \dot{\mu} &= 0. \end{aligned} \quad (5)$$

In which, v_b and g_e are the robot velocity vector in the body coordinate and the gravity acceleration vector in the NED coordinate, respectively. Next, R_χ and C_b^e are determined as follows [29]:

$$\begin{aligned} R_\chi &= \begin{bmatrix} 1 & s_{\phi_i} t_{\theta_i} & c_{\phi_i} t_{\theta_i} \\ 0 & c_{\phi_i} & -s_{\phi_i} \\ 0 & s_{\phi_i} s_{\theta_i} & c_{\phi_i} s_{\theta_i} \end{bmatrix} \\ C_b^e &= \begin{bmatrix} c_{\theta_i} c_{\psi_i} & s_{\phi_i} s_{\theta_i} c_{\psi_i} - c_{\phi_i} s_{\psi_i} & c_{\phi_i} s_{\theta_i} c_{\psi_i} + s_{\phi_i} s_{\psi_i} \\ c_{\theta_i} s_{\psi_i} & s_{\phi_i} s_{\theta_i} s_{\psi_i} + c_{\phi_i} c_{\psi_i} & c_{\phi_i} s_{\theta_i} s_{\psi_i} - s_{\phi_i} c_{\psi_i} \\ -s_{\theta_i} & s_{\phi_i} c_{\theta_i} & c_{\phi_i} c_{\theta_i} \end{bmatrix}. \end{aligned} \quad (6)$$

In which, C_b^e indicates the body to NED transformation matrix, where c and s signify the *cosine* and *sine* terms. According to the assumptions in [30], μ is assumed to be constant, so its rate will always be zero. The error dynamic equations are obtained using the perturbation method as follow:

$$\dot{\tilde{r}}_e = \hat{C}_b^e \tilde{v}_b + \delta C_b^e \hat{v}_b = \hat{C}_b^e \tilde{v}_b + \hat{v}_e \times \tilde{\chi}$$

$$\begin{aligned} \dot{\tilde{v}}_b &= \begin{bmatrix} -\frac{1}{m} (\hat{\mu} \tilde{v}_x + \tilde{\mu} \hat{v}_x) \\ -\frac{1}{m} (\hat{\mu} \tilde{v}_y + \tilde{\mu} \hat{v}_y) \\ \varepsilon_{az} \end{bmatrix} - (\Omega_b \tilde{v}_b - \hat{V}_b \varepsilon_\omega) \\ &+ g \begin{bmatrix} 0 & -c_\theta & 0 \\ c_\phi c_\theta & -s_f s_\theta & 0 \\ -s_\phi c_\theta & -c_\phi s_\theta & 0 \end{bmatrix} \tilde{\chi} \\ \dot{\tilde{\chi}} &= -\hat{C}_b^e \varepsilon_\omega \\ \dot{\tilde{\mu}} &= 0. \end{aligned} \quad (7)$$

In this equation, “ $\sim$ ” indicates the state error and “ $\hat{\cdot}$ ” stands for the estimated state variables. $\hat{V}_b$ and Ω_b are, respectively, the skew-symmetric matrices of estimated velocity vector and gyro measurements vector. Hence, (7) is rewritten in a discrete state space form as follows:

$$\begin{aligned} \dot{\tilde{x}}(t_{k+1}) &= F(t_k) \tilde{x}(t_k) + G(t_k) \varepsilon_u(t_k) \\ F(t_k) &= I \\ &+ \begin{bmatrix} 0 & \hat{C}_b^e & [\hat{v}_e \times] & 0 \\ 0 & -\left(\frac{\hat{\mu}}{m} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \Omega_b\right) & G & -\frac{1}{m} \begin{bmatrix} \hat{v}_x \\ \hat{v}_y \\ 0 \end{bmatrix} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \delta t \end{aligned} \quad (8)$$

$$\begin{aligned} G &= g \begin{bmatrix} 0 & -c_\theta & 0 \\ c_\phi c_\theta & -s_f s_\theta & 0 \\ -s_\phi c_\theta & -c_\phi s_\theta & 0 \end{bmatrix} \\ G(t_k) &= \begin{bmatrix} 0 & 0 \\ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \delta t & [\hat{v}_b \times] \delta t \\ 0 & -\hat{C}_b^e \delta t \\ 0 & 0 \end{bmatrix} \end{aligned} \quad (9)$$

where I is a 10×10 identity matrix, $[\hat{v}_b \times]$ is a skew-symmetric matrix of the body velocity vector, and δt denotes the timespan of the prediction step. To propagate the error covariance matrix, the system noise covariance matrix is

calculated as follows:

$$Q_d(t_k) = G(t_k) E \{ \varepsilon_u \varepsilon_u^T \} G^T(t_k)$$

$$E \{ \varepsilon_u \varepsilon_u^T \} = \begin{bmatrix} \sigma_a^2 & 0 & 0 & 0 \\ 0 & \sigma_\omega^2 & 0 & 0 \\ 0 & 0 & \sigma_\omega^2 & 0 \\ 0 & 0 & 0 & \sigma_\omega^2 \end{bmatrix}. \quad (10)$$

As mentioned earlier, in this method, the acceleration components on the rotor plane of one of the members are used as observations in the update step. Then, the observation equation is modeled as follow:

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} -\frac{\mu}{m} v_x \\ -\frac{\mu}{m} v_y \end{bmatrix}. \quad (11)$$

By linearizing (11), the model of observation error appears as follows:

$$\tilde{z}_d = H_d \tilde{x} + n_d$$

$$\begin{bmatrix} \tilde{z}_1 \\ \tilde{z}_2 \end{bmatrix} = -\frac{1}{m} \begin{bmatrix} \hat{\mu} \tilde{v}_x + \tilde{\mu} \hat{v}_x \\ \hat{\mu} \tilde{v}_y + \tilde{\mu} \hat{v}_y \end{bmatrix}$$

$$= -\frac{1}{m} \begin{bmatrix} 0 & 0 & 0 & \hat{\mu} & 0 & 0 & 0 & 0 & \hat{v}_x \\ 0 & 0 & 0 & 0 & \hat{\mu} & 0 & 0 & 0 & \hat{v}_y \end{bmatrix} \tilde{x} + n_d$$

$$R = E [n_d n_d^T] = \begin{bmatrix} \sigma_a^2 & 0 \\ 0 & \sigma_a^2 \end{bmatrix}. \quad (12)$$

Since the velocity elements are merely observed, the position vector is not observable as well. Therefore, if we do not have aiding information to correct the altitude error, this error will increase during the flight. However, since v_x and v_y are directly observed, the two-dimensional (2-D) position error obtained from integrating the velocity elements is expected to increase at a slower rate. The roll and pitch angles error are observable due to their effect on the differential equations of body velocity error. According to (13), this effect is due to the term of the gravity vector in the body coordinate in (7)

$$g \begin{bmatrix} 0 & -c_\theta & 0 \\ c_\phi c_\theta & -s_\phi s_\theta & 0 \\ -s_\phi c_\theta & -c_\phi s_\theta & 0 \end{bmatrix} \tilde{\chi} = g \begin{bmatrix} -\tilde{\theta} C_\theta \\ \tilde{\phi} C_\phi C_\theta - \tilde{\theta} S_\phi S_\theta \\ -\tilde{\phi} S_\phi C_\theta - \tilde{\theta} C_\phi S_\theta \end{bmatrix}. \quad (13)$$

In the presented method, the observations of accelerations involve a linear function of v_x and v_y ; hence, their error is well corrected. Due to the centrifugal acceleration term $\omega \times v_b$ in the differential equation of body velocity, v_z can also be observable in limited maneuvers. Therefore, by observing v_x and v_y , using a_x and a_y , v_z also is conditionally observed. The initial value of μ has an error and is needed to be corrected during the flight. The error of μ must be corrected swiftly, otherwise, it can be difficult to accurately estimate the other state variables. This happens at the beginning of the missions.

B. Multiagent Cooperative Navigation

In the following, the error propagation equation for the augmented system, including the error dynamic model of

all agents, can be written in a unit system as follows:

$$\begin{bmatrix} \tilde{X}_1 \\ \vdots \\ \tilde{X}_N \end{bmatrix}_{t_{k+1}} = \begin{bmatrix} F_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & F_N \end{bmatrix} \begin{bmatrix} \tilde{X}_1 \\ \vdots \\ \tilde{X}_N \end{bmatrix}_{t_k} + \begin{bmatrix} G_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & G_N \end{bmatrix} \times \begin{bmatrix} U_1 \\ \vdots \\ U_N \end{bmatrix}_{t_k} \quad (14)$$

Thus, the error covariance can be propagated through the following equation:

$$P(t_{k+1}^-) = F(t_k) P(t_k^+) F(t_k) + Q_d(t_k) \quad (15)$$

where $P(t_k^+)$ and $P(t_{k+1}^-)$ are, respectively, the prior corrected and the currently updated covariance matrices of the whole system that is defined as

$$P(t_{k+1}^-) = \begin{bmatrix} P_{11}(t_{k+1}^-) & \cdots & P_{1N}(t_{k+1}^-) \\ \vdots & \ddots & \vdots \\ P_{N1}(t_{k+1}^-) & \cdots & P_{NN}(t_{k+1}^-) \end{bmatrix}. \quad (16)$$

Assuming that each quadrotor is equipped with a UWB module (or can be a visual one), to measure the relative range and bearing angles, the measurement equation between the i th and j th agents can be written as follows:

$$\rho_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2} \quad (17)$$

$$\beta_{ij} = \tan^{-1} \frac{y_j - y_i}{x_j - x_i} - \psi_i \quad (18)$$

$$\alpha_{ij} = \tan^{-1} \frac{z_j - z_i}{d_{ij}} - \theta_i$$

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (19)$$

where ρ_{ij} , β and α , respectively, indicate the relative range, azimuth, and elevation angles between the agents. Then, the linearized observation equation is attained as follows:

$$\tilde{z}_{ij}(t_k) = H_{ij}(t_{k+1}) \tilde{X}(t_k) + \tilde{n}_{ij} \quad (20)$$

which $\tilde{n}_{ij}$ signifies the measurement zero-mean white noise with the covariance of

$$E [n_{ij} n_{ij}^T] = R_{ij}. \quad (21)$$

Similarly H_{ij} is formed as follows:

$$H_{ij} = \begin{bmatrix} 0 \cdots 0 & H_{ij}^i & 0 \cdots 0 & H_{ij}^j & 0 \cdots 0 \end{bmatrix} \quad (22)$$

$$H_{ij}^i = \begin{bmatrix} \frac{\partial \rho_{ij}}{\partial \tilde{X}_i} & \frac{\partial \beta_{ij}}{\partial \tilde{X}_i} & \frac{\partial \alpha_{ij}}{\partial \tilde{X}_i} \end{bmatrix}^T, H_{ij}^j = \begin{bmatrix} \frac{\partial \rho_{ij}}{\partial \tilde{X}_j} & \frac{\partial \beta_{ij}}{\partial \tilde{X}_j} & \frac{\partial \alpha_{ij}}{\partial \tilde{X}_j} \end{bmatrix} \quad (23)$$

where all derivatives are calculated as

$$\frac{\partial \rho_{ij}}{\partial \tilde{X}_i} = \begin{bmatrix} -\frac{\Delta \hat{x}_{ij}}{\hat{\rho}_{ij}} & -\frac{\Delta \hat{y}_{ij}}{\hat{\rho}_{ij}} & -\frac{\Delta \hat{z}_{ij}}{\hat{\rho}_{ij}} & 0_{1 \times 6} \end{bmatrix}$$

$$\frac{\partial \rho_{ij}}{\partial \tilde{X}_j} = \begin{bmatrix} \frac{\Delta \hat{x}_{ij}}{\hat{\rho}_{ij}} & \frac{\Delta \hat{y}_{ij}}{\hat{\rho}_{ij}} & \frac{\Delta \hat{z}_{ij}}{\hat{\rho}_{ij}} & 0_{1 \times 6} \end{bmatrix}. \quad (24)$$

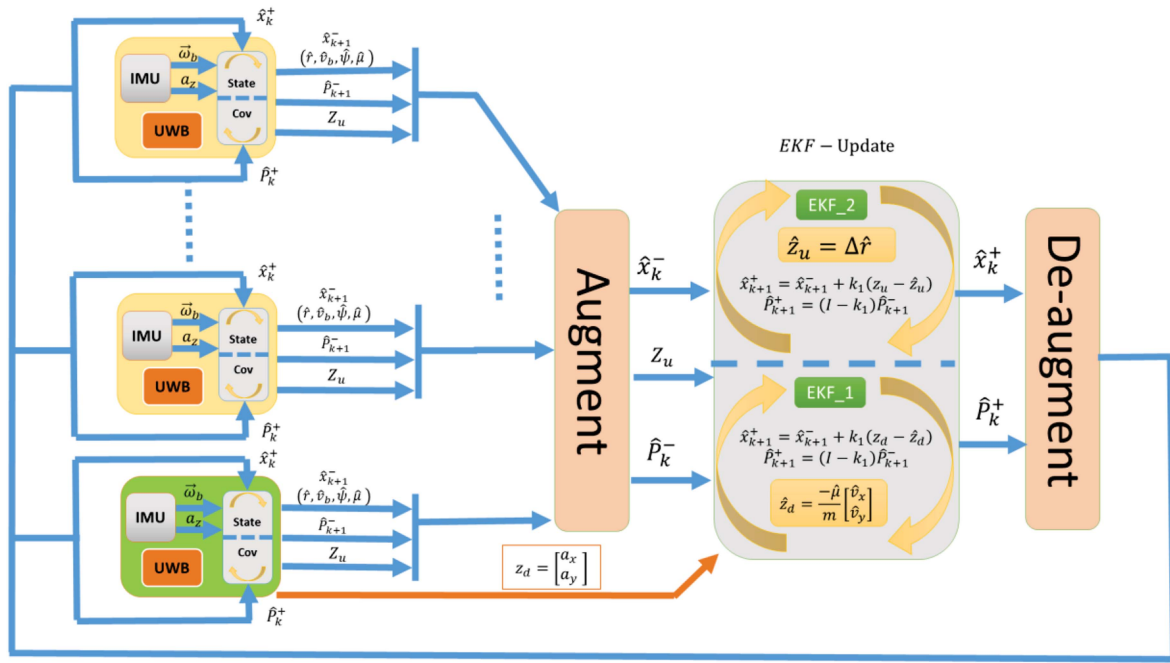


Fig. 3. Block diagram of DFB/MB-CN.

In which $\Delta \hat{x}_{ij} = \hat{x}_j - \hat{x}_i$, $\Delta \hat{y}_{ij} = \hat{y}_j - \hat{y}_i$ and $\Delta \hat{z}_{ij} = \hat{z}_j - \hat{z}_i$

$$\frac{\partial \beta_{ij}}{\partial \tilde{X}_i} = \begin{bmatrix} \frac{\Delta \hat{y}_{ij}}{d_{ij}^2} & -\frac{\Delta \hat{x}_{ij}}{d_{ij}^2} & 0_{1 \times 6} & -1 \end{bmatrix} \frac{\partial \beta_{ij}}{\partial \tilde{X}_j} = \begin{bmatrix} -\frac{\Delta \hat{y}_{ij}}{d_{ij}^2} & \frac{\Delta \hat{x}_{ij}}{d_{ij}^2} & 0_{1 \times 7} \end{bmatrix} \quad (25)$$

$$\begin{aligned} \frac{\partial \alpha_{ij}}{\partial \tilde{X}_i} &= \begin{bmatrix} \frac{\Delta \hat{x}_{ij} \Delta \hat{z}_{ij}}{d_{ij}^2 \cdot \rho_{ij}^2} & \frac{\Delta \hat{y}_{ij} \Delta \hat{z}_{ij}}{d_{ij}^2 \cdot \rho_{ij}^2} & -\frac{\hat{z}_{ij}}{\rho_{ij}^2} & 0_{1 \times 4} & -1 & 0 \end{bmatrix} \\ \frac{\partial \alpha_{ij}}{\partial \tilde{X}_j} &= \begin{bmatrix} -\frac{\Delta \hat{x}_{ij} \Delta \hat{z}_{ij}}{d_{ij}^2 \cdot \rho_{ij}^2} & -\frac{\Delta \hat{y}_{ij} \Delta \hat{z}_{ij}}{d_{ij}^2 \cdot \rho_{ij}^2} & \frac{\hat{z}_{ij}}{\rho_{ij}^2} & 0_{1 \times 6} \end{bmatrix}. \end{aligned} \quad (26)$$

Regarding the observability analysis of the proposed model, due to relative measurements, references such as [7]–[9], and [13] give more detailed information for the interested readers.

C. MB-CN Structure

In the following, the introduced MB method (see Fig. 2) is integrated with the CN algorithm. The proposed method, instead of the conventional IMU-based DR, is based on (5)–(14), while each agent separately propagates its model. Therefore, the μ of each agent is augmented to the state vector of the whole system. Moreover, in addition to relative position observations between the agents, the body acceleration measurements, a_x and a_y of one of the agents, is also used as an observation in the update step. Fig. 3 presents the related block diagram of an N-agent MB-CN. In the proposed method, the propagations are accomplished in each agent distinctly. Likewise, the update step is performed on a single central EKF, which can be implemented on a ground station or the processor of one agent. Each quadrotor can be equipped with IMU and UWB device for measuring

and communicating the relative position data, including relative distance and bearing angle between the agents.

The update step has two separate sections, each of which performs the update at a different rate. In the first part, the update is performed at a high rate using acceleration observations of only one agent of the group. In the second section, the update is performed at a lower rate using relative position observations generated by UWB transceivers. Only in one agent of the group, the measured accelerations a_x and a_y , is added to the output vector. Earlier than the update step starts, all data are aggregated into a unified structure. After the update step, the estimated data including the states and covariances are distributed among the agents. In CN, each relative observation between agents forms a cross-correlation between them. When the agents observe each other, all elements of the covariance matrix, including the cross-correlations and covariances of each agent, gains a nonzero value. Hence, it is necessary that the a_x and a_y measured in one of the agents, is augmented to the observation vector of the whole system.

III. RESULTS: SIMULATION AND PERFORMANCE EVALUATION

This section is dedicated to the performance evaluation of the proposed method by performing several simulations alongside the demonstration of the related metrics. The simulation platform is developed under the MATLAB environment. Here the first agent provides acceleration observations required for the update step. The rate of update using relative observations is 2 Hz while for the case of acceleration observations it is allotted 20 Hz. During the simulations, each agent flies through a hexagonal path at a different altitude. In the following, Table I declares the specified value

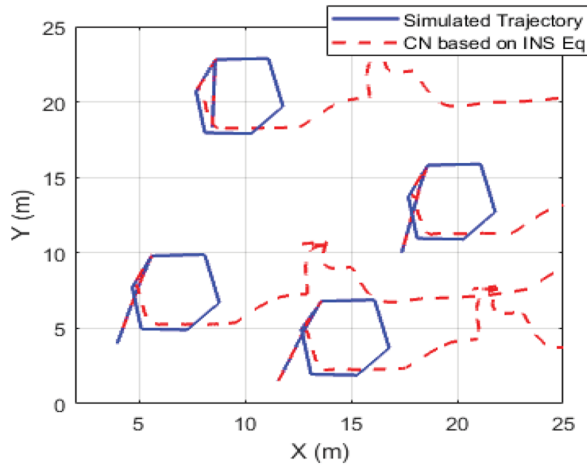
TABLE I
Simulation Parameters and Values

PARAMETER NAME	SYMBOL	PARAMETER VALUE
Mass	m	1.023 kg
Inertia Tensor	J	$\begin{bmatrix} 0.0095 & 0 & 0 \\ 0 & 0.0095 & 0 \\ 0 & 0 & 0.0186 \end{bmatrix}$ kg.m ²
Thrust coefficient	C_T	1.4865×10^{-7}
Rotor torque Coefficient	C_τ	2.9250×10^{-9}
Rotor drag force Coefficient	μ	0.1023
Gyro std	σ_ω	100 ^{deg} /hr
Accelerometer std	σ_a	1.2×10^{-2} m/s ²
IMU output rate	-	20 Hz
UWB distance std	σ_d	1 cm
UWB angle std	$\sigma_\alpha, \sigma_\beta$	2 Deg
UWB output rate	-	2 Hz
UWB distance measure limit	-	15 m

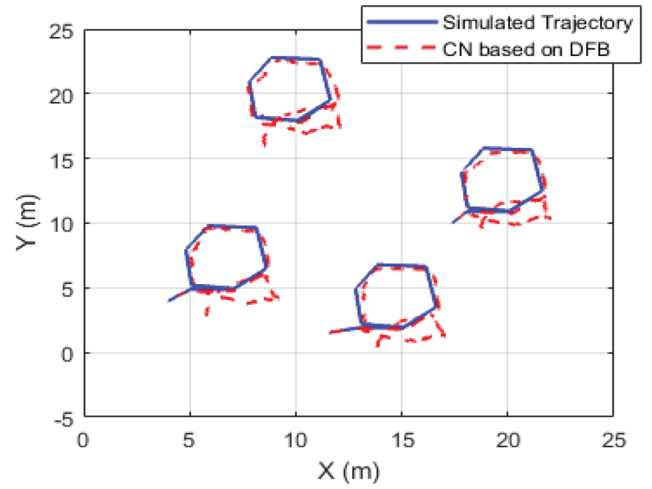
**std: standard deviation.



Fig. 4. DR error based on DFB and IMU propagations in the 100 s.



(a)



(b)

Fig. 5. CN based on two different dynamic equations. (a) Conventional INS. (b) DFB.

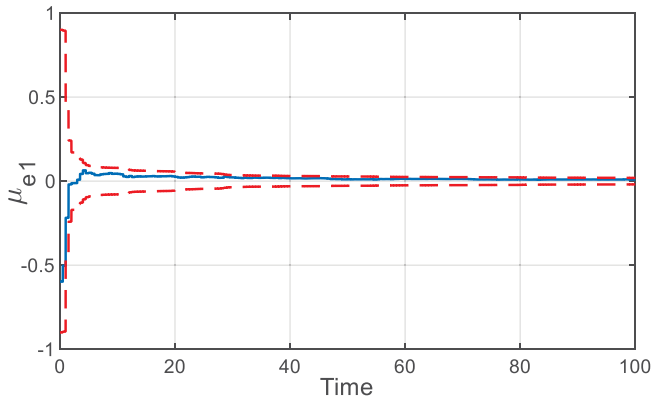


Fig. 6. Estimated μ for the first agent.

of the related variables in the simulation toolset. For more realism, in this article, an integrated waypoint navigation (following) system is developed to handle the guidance commands; which encompasses an attitude control system as its inner loop and a position control system as its outer loop. In the following, first, the CN method based on the new dynamic equation (DFB: drag-force-based) is evaluated. Next, the MB-CN performance evaluations are performed via the simulation environment. Finally, the efficiency of the proposed method is assessed through performance indices.

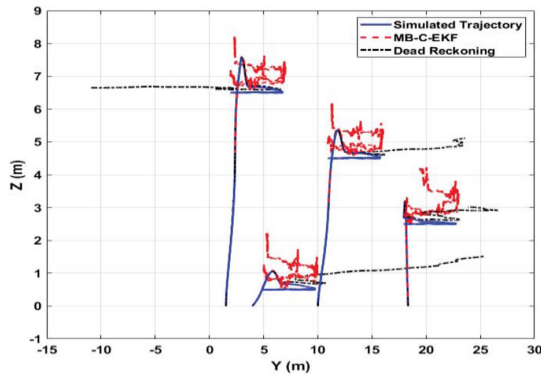
A. Evaluation of DFB-CN Method

To assess the effect of the DFB dynamic equations on the CN, two series of simulations are performed. In the first one, the CN is based on the new equations (DFB). Then, the evaluation process of the CN, based on conventional IMU, is conducted. In both simulations, only relative observations are used in the update step. Since the simulation conditions are the same, the performance of both methods can be examined and compared with each other. In the following, at first, only the DR navigation,

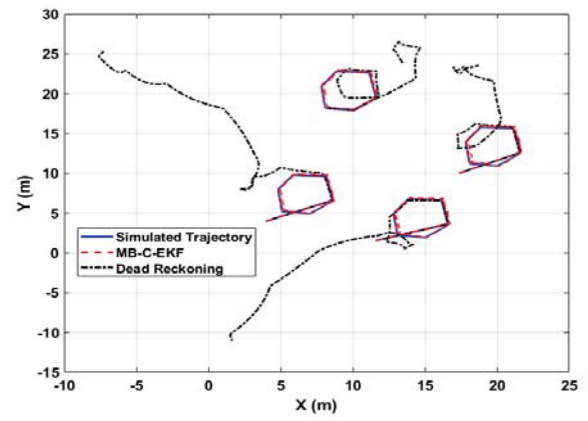
based on these two dynamic equations is simulated without any update. In this simulation, the DR navigation code executes 200 times where every flight lasts 100 s. In all simulations, the quadrotor is simulated in a stationary state on the ground. Here, the parameter μ is known exactly just before the simulation starts. In this regard, Fig. 4 shows the 2-D position error in the horizontal plane in the 100 s versus the simulation run number. As depicted in Fig. 4, the dispersion of the position error regarding the IMU case is greater than the one in the DFB instance. One reason is that in the IMU case, the noise of all accelerometers and gyros is propagated, while in DFB equations, only the noise of one accelerometer and three gyros are propagated. It should be noted that the dispersion of the altitude errors are the same in both cases; since the errors initiate from the vertical measurements a_z . Fig. 5 depicts the estimated 2-D positions in the horizontal plane for both cases. This figure also indicates that based on the DFB dynamic equations, the CN experiences higher performance. Again, It should be noted that only relative observations are employed here and acceleration observations are not used for updates. One of the reasons for the performance improvement is that the estimation algorithm which uses DFB equations involves a higher capacity to estimate v_x and v_y and consequently the roll and pitch angles as well. One of the remarkable results shown in Fig. 6 is the improvement of the μ estimation errors.

B. Evaluation of MB-CN Method

In this section, the simulation results of the MB-CN, for a four-agent group of aerial robots, are represented. Here, the simulation results of two agents of the group are portrayed, one of them is the agent that provides the acceleration observations required for the update step. Fig. 7(a) and (b) exhibits the 2-D trajectory of each agent on the vertical plane (Z-Y) and horizontal plane (X-Y), respectively. The figures display the simulated trajectory (as ground truth) by

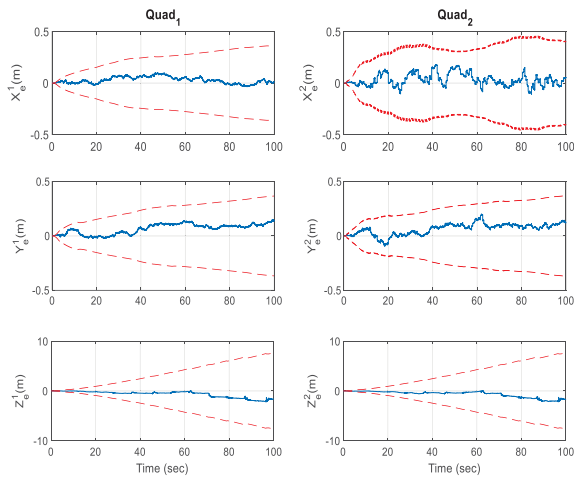


(a)

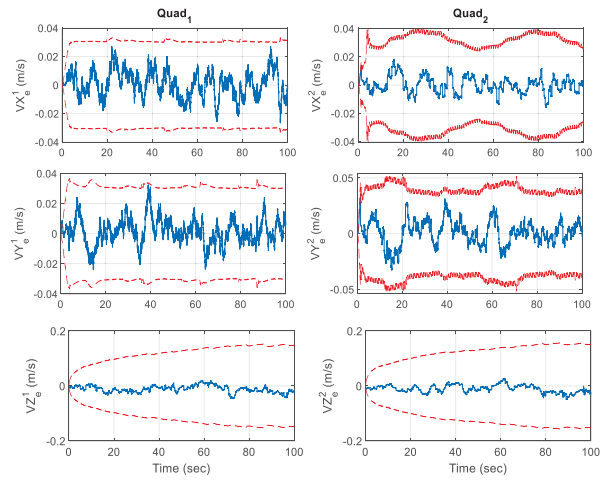


(b)

Fig. 7. Position of a four-agent group of quadrotors in (a) vertical plane and (b) horizontal plane.



(a)



(b)

Fig. 8. Quadrotors 1 and 2, $\pm 3\sigma$ (a) position error, and (b) velocity error.

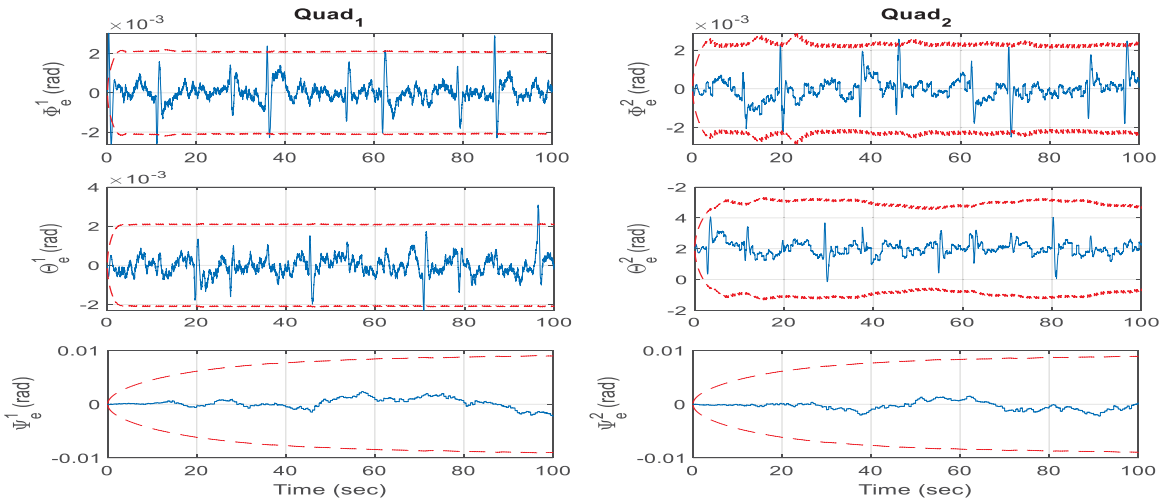


Fig. 9. $\pm 3\sigma$ attitude error for quadrotors 1 and 2.

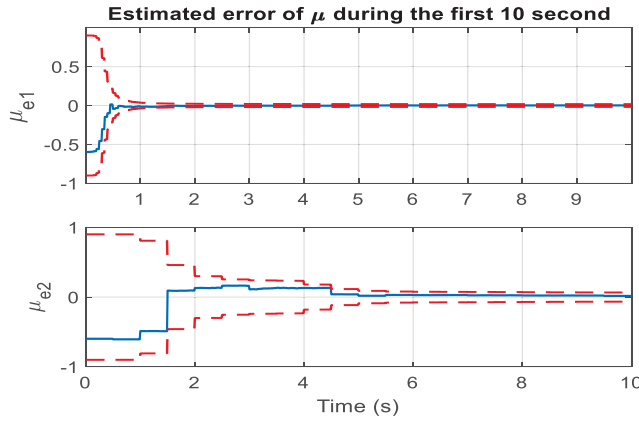


Fig. 10. Each agent's $\pm 3\sigma$ estimation error of μ .

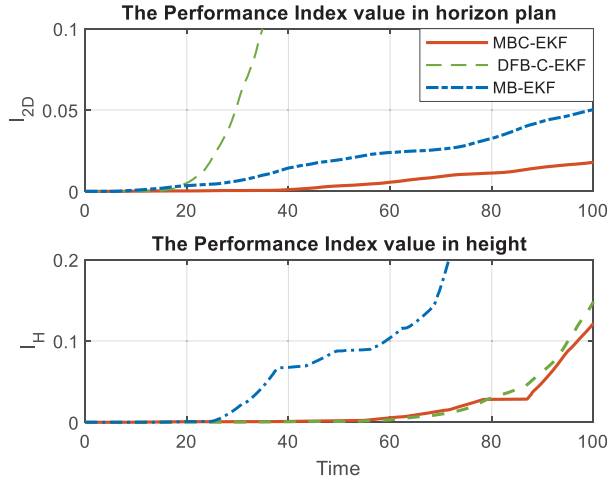


Fig. 11. Index value versus time for three introduced method.

a blue line, the trajectory of the model-based cooperative EKF (MBC-EKF) with the red dashed line, and the DR trajectory with the black dotted line. Here, the DR equations are the same as the conventional inertial navigation equations; As expected, the position predictions during the flight are well estimated. The same as before, no information source of the absolute position data is employed during the evaluation process; Therefore, the position error is expected to increase over time. One of the properties of the proposed method (MB-CN) compared to the original one (CN) is that the growth rate of position errors and its covariance is significantly reduced (see Fig. 8). Whereas in the CN method, in which only relative observations are used, the position of the agents diverges after a short time. In the proposed method, due to the direct observation of v_x and v_y of only one agent of the group (by accelerometers), the error values of v_x , v_y , φ , and θ angles of the other agents are also well updated and their error covariance is converged rapidly. Figs. 8 and 9 reveal the prescribed observations. In the following, Fig. 10 shows the estimation error of μ in quadrotor 1 and 2 during the first ten seconds of flight. It is perceived that the estimation error and its covariance converge to zero after a while. In quadrotor 1, the μ error and its covariance are converged at a faster rate, because

it is directly observed in quadrotor 1 by the accelerometer measurements. The interesting point here is that the error of μ in the other agents is corrected by relative observations between the agents. Therefore, it is obvious that the convergence time of the μ error in other agents is higher than the first agent.

C. Index-Based Performance Evaluations

To evaluate the proposed method, its performance is compared with the original ones using two indices. The discussed methods include the following:

- 1) MBC-EKF (model-based cooperative EKF);
- 2) MB-EKF (model-based EKF);
- 3) DFB-C-EKF (drag-force-based cooperative EKF).

Based on the prior deliberations, MBC-EKF is a model-based cooperative method, MB-EKF is a model-based non-cooperative one, and DFB-C-EKF is a cooperative method, in which the equations are based on the drag force model instead of IMU propagations. To examine them, the first index is the total position error at each moment of flight time

$$I_{2-D}(t_n) = \frac{\sum_{i=1}^n \int_0^{t_n} (E_{2-D}^T E_{2-D})_i dt}{n \cdot \int_0^{t_n} dt}$$

$$E_{2-D}(t) = \begin{bmatrix} (x_i(t) - \hat{x}_i(t)) \\ (y_i(t) - \hat{y}_i(t)) \end{bmatrix}$$

$$I_H(t_n) = \frac{\sum_{i=1}^n \int_0^{t_n} (E_H^2)_i dt}{n \cdot \int_0^{t_n} dt}$$

$$E_H(t) = z_i(t) - \hat{z}_i(t) \quad (27)$$

where I_{2-D} is the value of the index introduced on the horizontal plane and I_H is the value of the index introduced in the height (altitude) channel, and t_n is the total flight time.

Another index that is commonly used in evaluations is the root-mean-square error

$$\text{RMSE}_{2-D} = \sqrt{\text{mean} \left(\frac{\sum_{i=1}^n (\bar{E}_X^2 + \bar{E}_Y^2)}{n} \right)}$$

$$\bar{E}_X = \begin{bmatrix} (x - \hat{x})_{t_0} \\ \vdots \\ (x - \hat{x})_{t_f} \end{bmatrix}, \bar{E}_Y = \begin{bmatrix} (y - \hat{y})_{t_0} \\ \vdots \\ (y - \hat{y})_{t_f} \end{bmatrix}$$

$$\text{RMSE}_H = \sqrt{\text{mean} \left(\frac{\sum_{i=1}^n \bar{E}_H^2}{n} \right)}$$

$$\bar{E}_H = \begin{bmatrix} (z - \hat{z})_{t_0} \\ \vdots \\ (z - \hat{z})_{t_f} \end{bmatrix} \quad (28)$$

where RMSE_{2-D} and RMSE_H are the root-mean-square error of the position in the horizontal plane and height channel, respectively. The three estimation methods introduced (MBC-EKF, MB-EKF, DFB-C-EKF) are simulated separately for 100 s. The value of the index introduced in (27) for each agent is shown in Fig. 11. The simulation

TABLE II
Root-Mean-Square Error of the Position Estimations

The estimation method	$RMSE_{2D}$	$RMSE_H$
MBC-EKF	0.2170	1.3880
MB-EKF	0.3654	5.2949
DFB-C-EKF	9.80	2.1095

conditions are the same for all three methods including flight path, initial values, and sensors specifications (see Table I).

The upper graph in Fig. 11, exhibits the position error index for the horizontal plane and the lower one presents that of the height channel. Based on the prescribed indices, Table II yields the corresponding index for each estimation method. Where $RMSE_{2-D}$ and $RMSE_H$ stand for the root-mean-square error of the position in the horizontal plane and height channel, respectively. These values are obtained for 200 runs of each method. It is also noticeable that the performance of MBC-EKF for both the position error indices on the horizontal plane and the height channel is superior to the other methods.

Furthermore, as can be perceived from the outcomes, specifically from Fig. 11, the values of the indices have an incremental trend in all three methods. This result is due to the lack of absolute position information such as GNSS signals. The value of the index of position error on the horizontal plane in DFB-C-EKF and that of the height error index in MB-EKF grow much faster. This is due to the weakness of the CN in estimating the position on the horizontal plane and the same for MB-EKF in the estimation of altitude. Therefore, it can be said that the combination of these two algorithms, which has led to a model-based cooperative navigation system, improves the weakness of both methods.

IV. CONCLUSION

In this article, an MB method was employed to enhance the performance of the CN of a group of quadrotors. This method is only for multirotor type robots because the dynamic model used in the proposed method is based on DFB; due to rotors drag, which is a linear function of v_x and v_y . Therefore, it is possible to directly observe body velocities in the same directions by the accelerometers data a_x and a_y . By observing the velocities, the roll and pitch angles can also be updated well. By transferring the velocities in the NED frame and integrating over time, the position on the horizontal plane can be obtained with appropriate accuracy. However, although there is no external source of absolute position data, the position errors and their covariance increase over time at a much slower rate relative to the CN. It was also shown in the simulations that the parameter μ in the drag force model of each quadrotor can be well estimated without acceleration observations via only relative observations. The results illustrate that RMSE of

MBC-EKF is reduced 97% in the 2-D (horizontal) position and 40% in the vertical one, relative to the DFB-C-EKF (CN based on DFB) algorithm; while it is 42% in the horizontal plane and 73% on the vertical one, relative to MB-EKF. The followings are some activities that can be conducted as further research:

- 1) sensitivity analysis of the proposed method for various parameters such as drag coefficients, mass and inertia tensor changes, and also measurement limits (on the range and bearing measurements);
- 2) using of experimental and test data to validate the proposed method;
- 3) development of a model-based CN for a nonhomogeneous group;
- 4) development and implementation of a model-based SLAM.

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